VLAN CONCEPT

WHY VLAN ???

Lets imagine that for every broadcast the switch will make multiple copies and send its to every port and so it creates very high no of frames flowing through the network and so if now any PC want to use network then the network is already having very large traffic.so such a high no. of frames present in a network causes :

- ***Degradation of switch performance***
- ***Degradation of PC`s performance***
- ***Degradation of useful Bandwidth of a link***

so basically VLAN`s ARE USED TO CONTROL THE BROADCAST DOMAIN.

WHAT IS VLAN

VLAN is a logical grouping of networking devices.

VLANs are used to connect different networks.

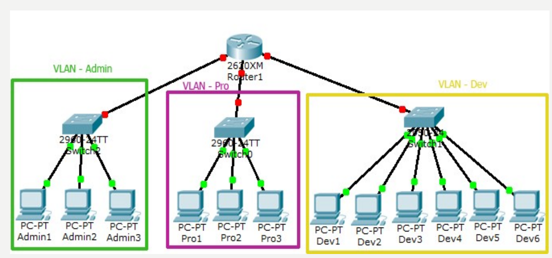
When we create VLAN, we actually break large broadcast domain in smaller broadcast domains. Consider VLAN as a subnet.

Same as two different subnets cannot communicate with each other without router, different VLANs also requires router to communicate.

It means that the devices in the same VLAN may be widely separated in the network, both by geography and location.

VLANs logically segment the network into different broadcast domains so that packets are only switched between ports that are designated for the same VLAN.

- When we define a VLAN on a switch at that time we are actually dividing the switch in to the smaller swich virtually where these virtual switchs are working in the different network (subnet).



- VLANs logically divide a switch into multiple, independent switches at Layer 2.
- Each VLAN is its own broadcast domain.
- Each VLAN should be in its own subnet.

VLAN helps you group users together according to their function rather than their physical location. This means you can use the same network for hosts in different floors (of course they can communicate with each other).

ADVANTAGES OF VLAN

VLAN provides following advantages

- **Solve broadcast problem :**

  Each VLAN has a separate broadcast domain. Logically VLANs are also subnets. Each VLAN requires a unique network number known as VLAN ID. Devices with same VLAN ID are the members of same broadcast domain and receive all broadcasts.

- **Reduce the size of broadcast domains :**

  VLAN increase the numbers of broadcast domain while reducing their size.

  Thus more VLAN means more broadcast domain with less devices.

- **Allow us to add additional layer of security :**

VLANs enhance the network security. In a typical layer 2 network, all users can see all devices by default. Any user can see network broadcast and responds to it.

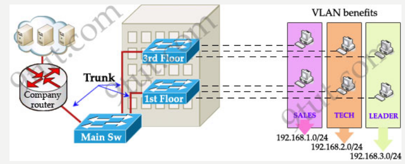
This could create real trouble on security platform. Properly configured VLANs gives us total control over each port and users. With VLANs, you can control the users from gaining unwanted access over the resources. We can put the group of users that need high level security into their own VLAN so that users outside from VLAN can't communicate with them.

- **Make device management easier :**

Device management is easier with VLANs. Since VLANs are a logical approach, a device can be located anywhere in the switched network and still belong to the same broadcast domain. We can move a user from one switch to another switch in same network while keeping his original VLAN.

- **Allow us to implement the logical grouping of devices by function instead of location**

VLANs allow us to group the users by their function instead of their geographic locations. Switches maintain the integrity of your VLANs.



FEATURES OF VLAN

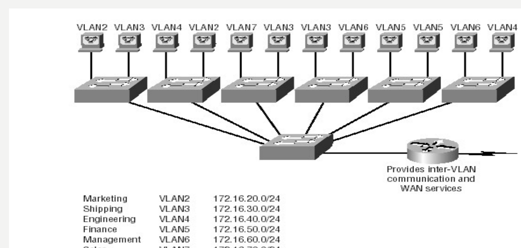
- Simplify network management
- Provides a level of security over a flat network
- Flexibility and Scalability

VLANs

Essentially create broadcast domains

- **Greatly reduces broadcast traffic**
- **Ability to add wanted users to a VLAN regardless of their physical location**
- **Additional VLANs can be created when network growth consumes more bandwidth**

VLANS REMOVE THE PHYSICAL BOUNDARY



VLAN MEMBERSHIP

VLAN membership can be assigned to a device by one of two methods

1. **Static :**

Assigning VLANs statically is the most common and secure method. It is pretty easy to set up and supervise. In this method we manually assign VLAN to switch port. VLANs configured in this way are usually known as port-based VLANs.

Static method is the most secure method also. As any switch port that we have assigned a VLAN will keep this association always unless we manually change it. It works really well in a networking environment where any user movement within the network needs to be controlled.

2 . **Dynamic :**

In dynamic method, VLANs are assigned to port automatically depending on the connected device. In this method we have configure one switch from network as a server. Server contains device specific information like MAC address, IP address etc. This information is mapped with VLAN. Switch acting as server is known as VMPS (VLAN Membership Policy Server). Only high end switch can configured as VMPS. Low end switch works as client and retrieve VLAN information from VMPS.

Dynamic VLANs supports plug and play movability. For example if we move a PC from one port to another port, new switch port will automatically be configured to the VLAN which the user belongs. In static method we have to do this process manually.

VLAN CONNECTIONS

1. Access link

Access link connection is the connection where switch port is connected with a device that has a standardized Ethernet NIC. Standard NIC only understand IEEE 802.3 or Ethernet II frames.

Access link connection can only be assigned with single VLAN. That means all devices connected to this port will be in same broadcast domain.

A link that is part of only one VLAN

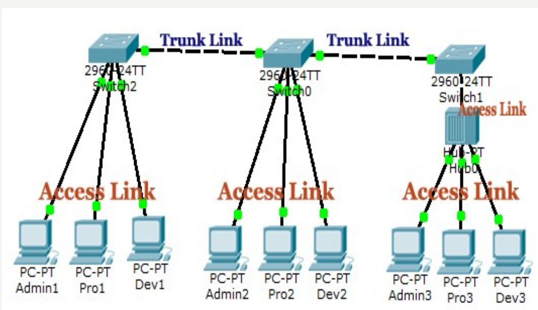
2 . Trunk link

Trunk link connection is the connection where switch port is connected with a device that is capable to understand multiple VLANs.

Usually trunk link connection is used to connect two switches or switch to router.

VLAN can span anywhere in network, that is happen due to trunk link connection.

Trunking allows us to send or receive VLAN information across the network. To support trunking, original Ethernet frame is modified to carry VLAN information.



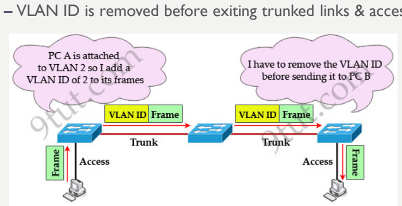
BASIC VLAN CONCEPTS

- **Port-based VLANs**
  - Each port on a switch is in one and only one VLAN (except trunk links)
- **Tagged Frames**
  - VLAN ID and Priority info is inserted (4 bytes)
- **Trunk Links**
  - Allow for multiple VLANs to cross one link
- **Access Links**
  - The edge of the network, where legacy devices attach
- **Hybrid Links**
  - Combo of Trunk and Access Links
- **VID**
  - VLAN Identifier

Frame Tagging

Definition: A means of keeping track of users & frames as they travel the switch fabric & VLANs

- User-defined ID assigned to each frame
- VLAN ID is removed before exiting trunked links & access links



THANK YOU